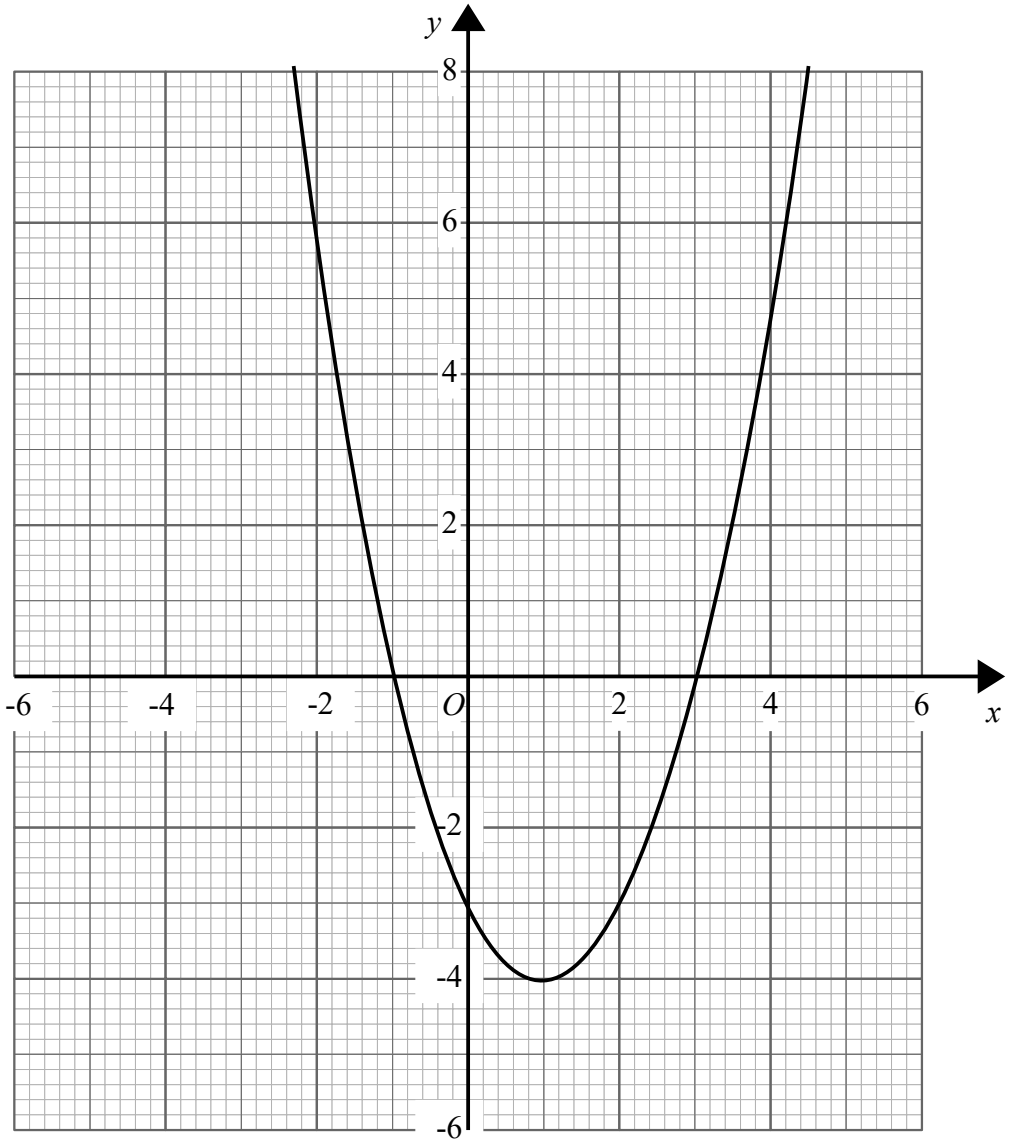
1 Here is the graph of $y = x^2 - 2x - 3$



(a) Write down the turning point of the graph $y = x^2 - 2x - 3$

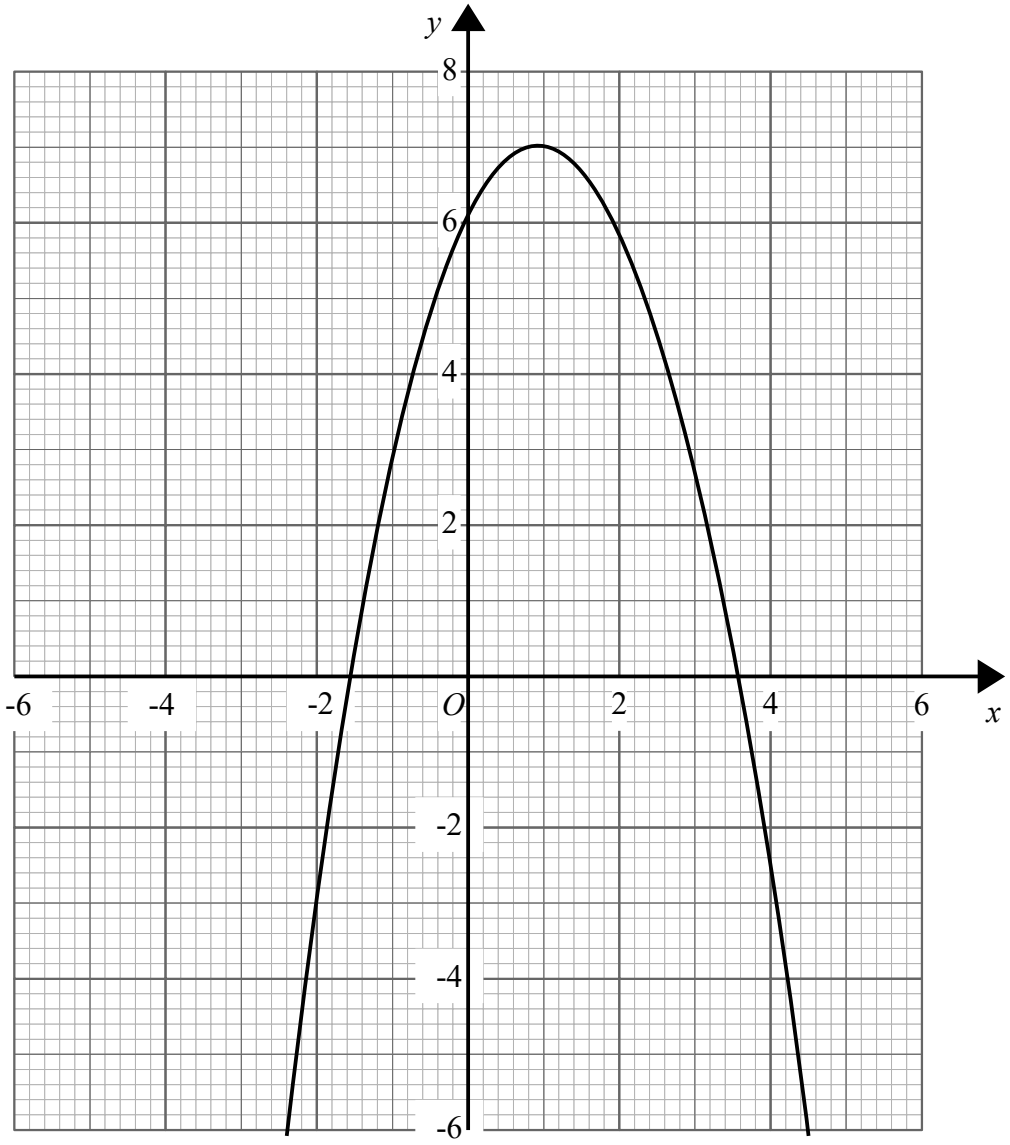
(……..…… , ……....….)
(1)

(b) Use the graph to find the roots of the equation $x^2 - 2x - 3 = 0$

….....….....….....…
(2)

(Total for question 1 is 3 marks)

2 Here is the graph of $y = 2x + 6 - x^2$



(a) Write down the turning point of the graph $y = 2x + 6 - x^2$

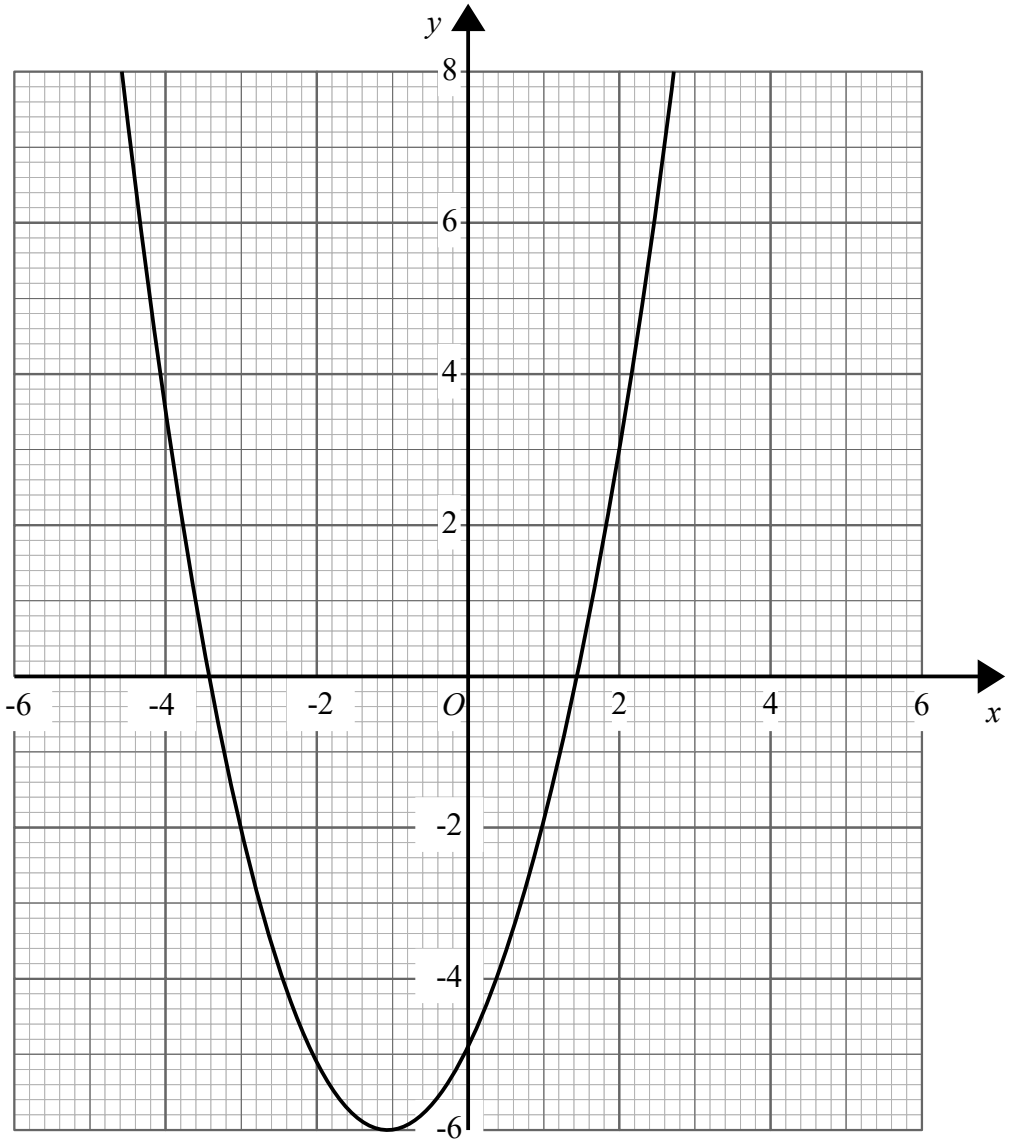
(……..…… , ……....….)
(1)

(b) Use the graph to find the roots of the equation $x^2 = 2x + 6$

….....….....….....…
(2)

(Total for question 2 is 3 marks)

3 Here is the graph of $y = x^2 + 2x - 5$



(a) Write down the turning point of the graph $y = x^2 + 2x - 5$

(……..…… , ……....….)
(1)

(b) Use the graph to find the roots of the equation $x^2 + 2x - 5 = 2$

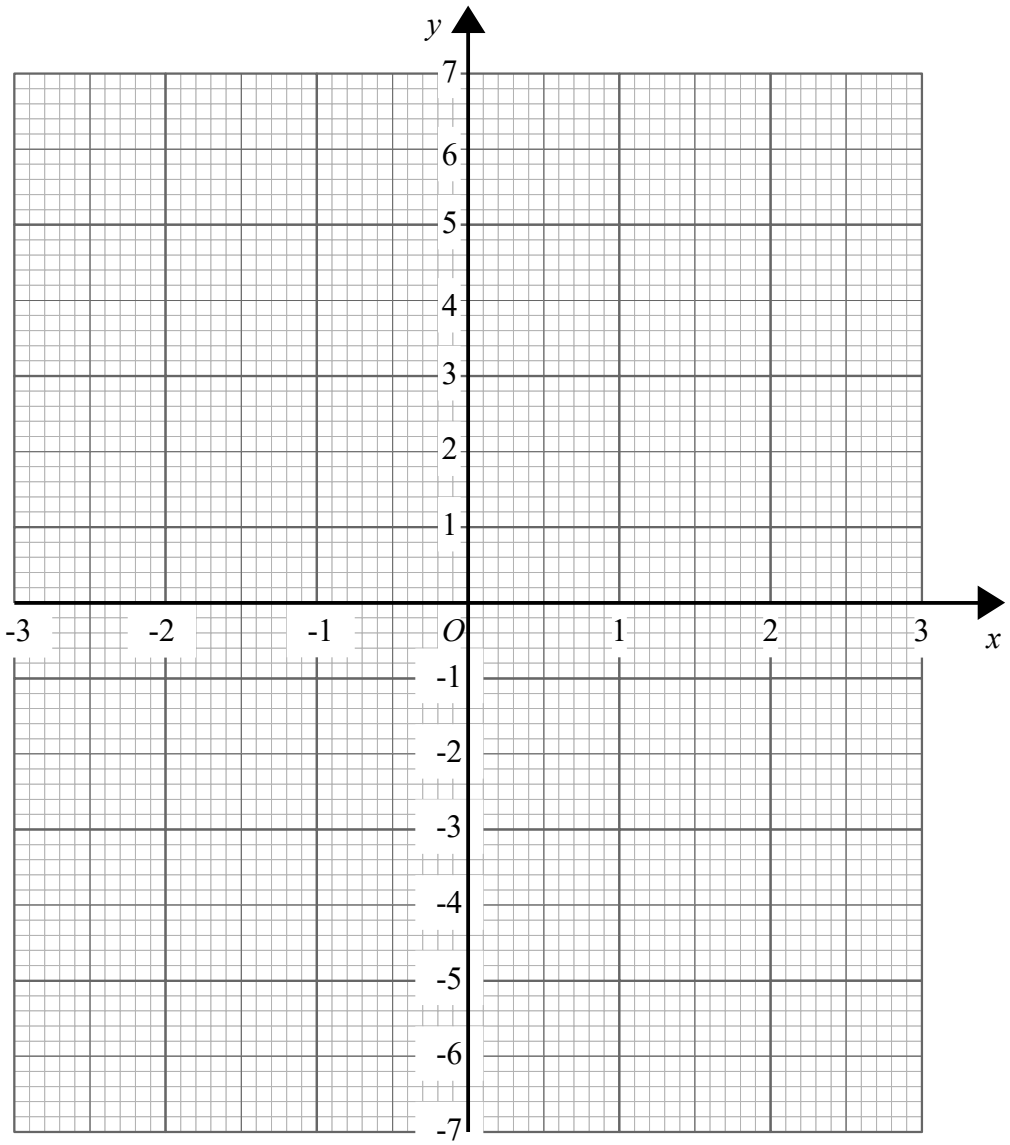
….....….....….....…
(2)

(Total for question 3 is 3 marks)

4 Complete the table of values for $y = x^2 + x - 6$

x	-3	-2	-1	0	1	2	3
y				-6		0	

(2)



(a) On the grid draw the graph of $y = x^2 + x - 6$ for values of x from –3 to 3 (2)

(b) Use the graph to find estimates of the solutions to the equation $x^2 + x - 6 = -2$

…….....…….....…….....…

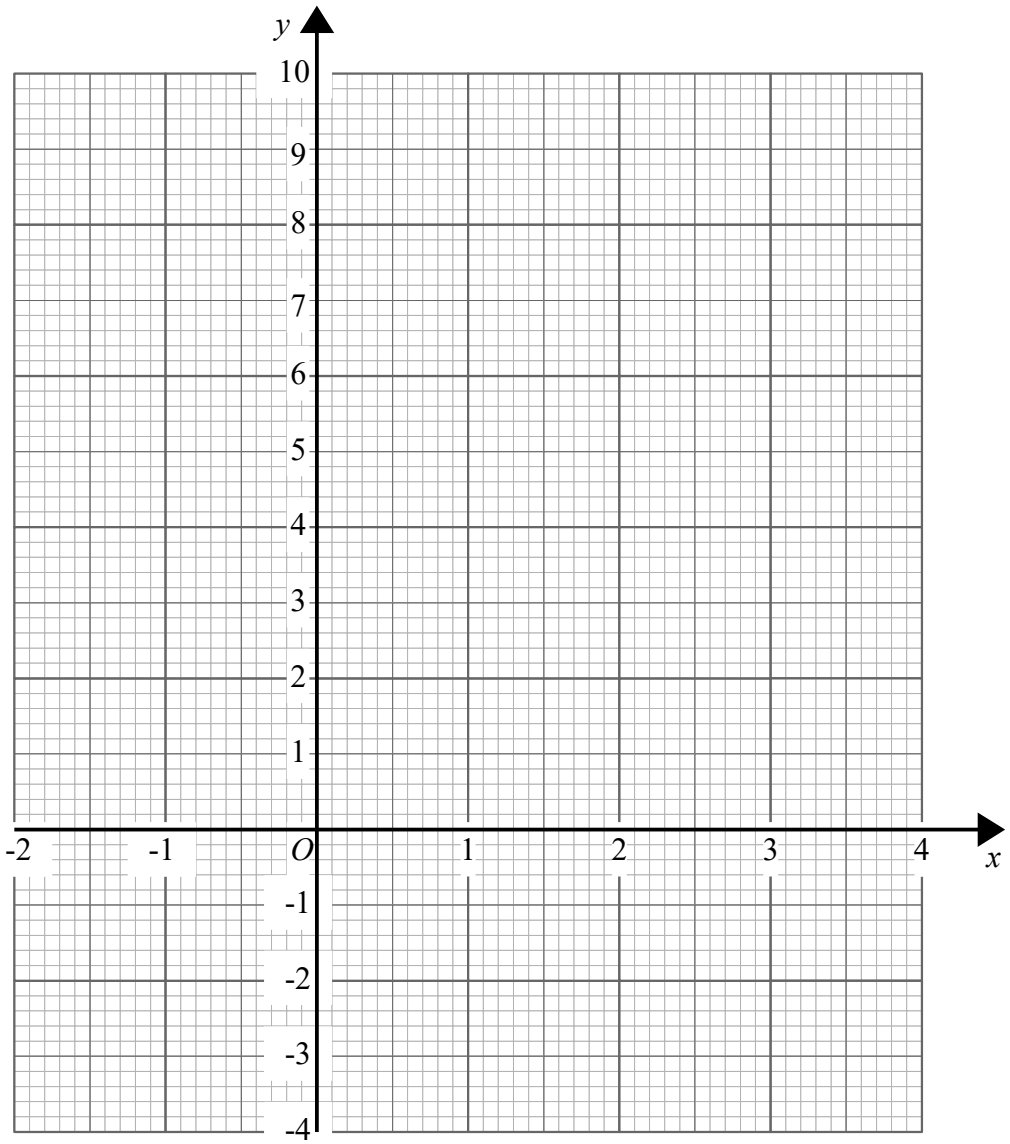
(2)

(Total for question 4 is 6 marks)

5 Complete the table of values for $y = x^2 - 3x - 1$

x	-2	-1	0	1	2	3	4
y							

(2)



(a) On the grid draw the graph of $y = x^2 - 3x - 1$ for values of x from –2 to 4 (2)

(b) Use the graph to find an estimate of the turning point of the graph $y = x^2 - 3x - 1$

…….....…….....…….....…

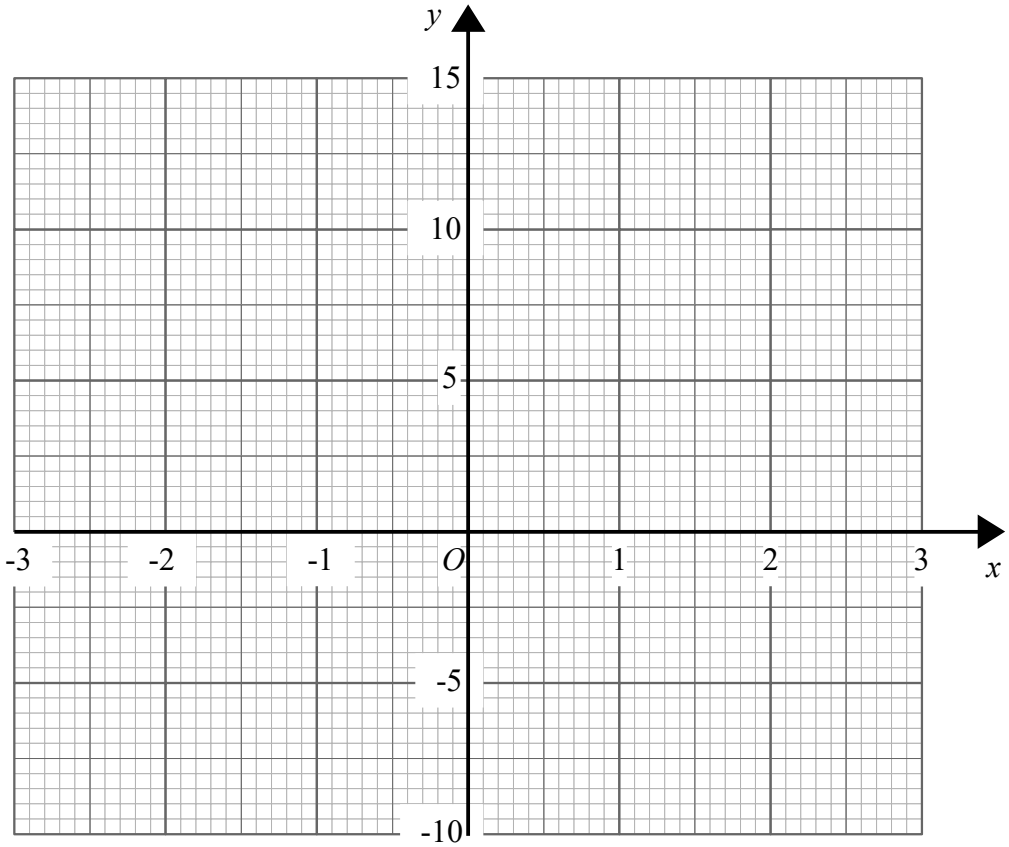
(2)

(Total for question 5 is 6 marks)

6 Complete the table of values for $y = x^2 - 2x - 5$

x	-3	-2	-1	0	1	2	3
y							

(2)



(a) On the grid draw the graph of $y = x^2 - 2x - 5$ for values of x from –3 to 3 (2)

(b) Use the graph to find an estimate of a solution to the equation $x^2 = 2x + 5$

…….....…….....…….....…

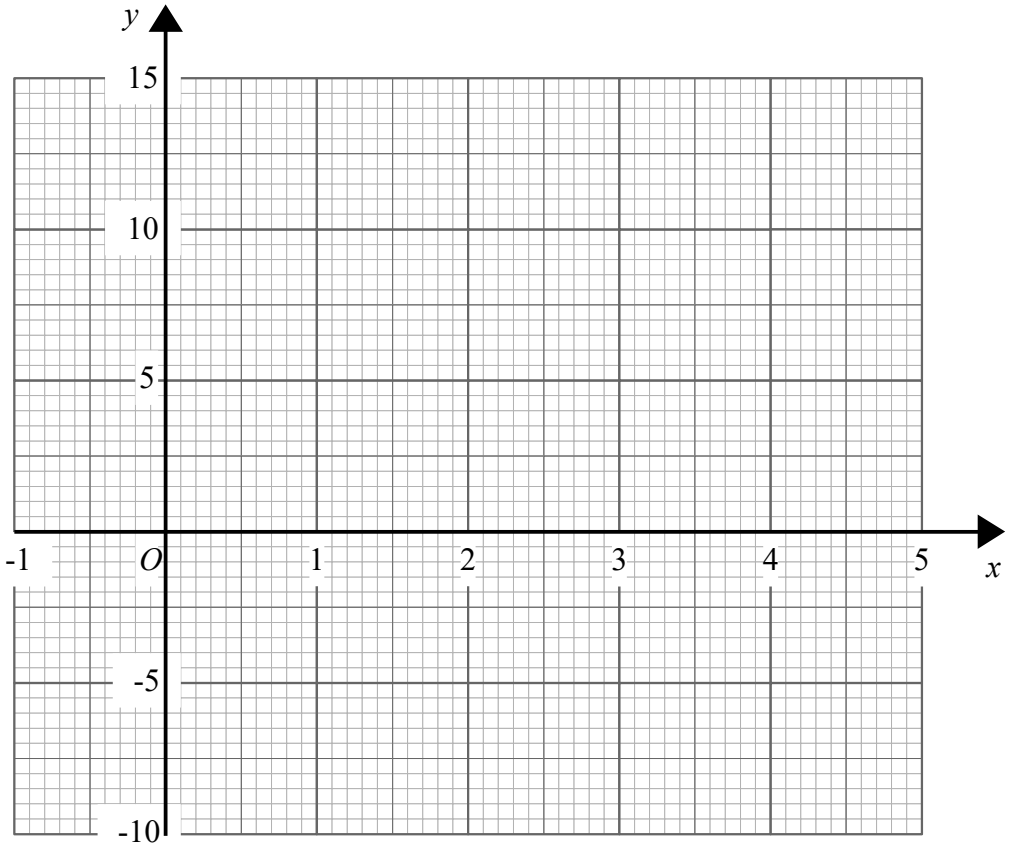
(1)

(Total for question 6 is 5 marks)

7 Complete the table of values for $y = 7x - x^2$

x	-1	0	1	2	3	4	5
y							

(2)



(a) On the grid draw the graph of $y = 7x - x^2$ for values of x from –1 to 5 (2)

(b) Use the graph to find an estimate of the turning point of the graph $y = 7x - x^2$

…….....…….....…….....…

(2)

(c) Find the solutions to the equation $7x - x^2 = 0$

…….....…….....…….....…

(2)

(Total for question 7 is 8 marks)